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DBMS (Database Management System)

Assignment No. 2

Ques 1 :→ Fill in the blanks.

a) The ER Diagram visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

b) In the Relational Model, data is organized into tables, with relationships between entities defined through foreign key constraints.

c) The Object-Oriented Model treats, data as a collection of objects, each with unique attributes and methods, mirroring real-world entities and interactions.

d) Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

Ques 2 → In a healthcare system, discuss how data abstraction can simplify the interaction between healthcare professionals and patient records.

Ans → Data Abstraction involves hiding the complexity and internal details of database and only exposing essential details to the users. It allows users to interact with database without needing to understand underlying details of the database. In healthcare system data abstraction easy the interaction b/w healthcare professionals and patient records.

a) → Simplified Access to information :-

Doctors and nurses do not need to understand the complex database structure. They can simply view patient details such as name, age, diagnosis and treatment through an easy interface.

b) → Different views for Different users :-

Different healthcare professionals require different info.

→ Doctors see medical records and prescriptions.

→ Nurses see medication schedules and patient status.

→ Administrative staffs see billing details.

c) → Improved data security :-

Sensitive patient data is protected because users only see the information relevant to their role.

Ques 3 → Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Ans → Benefits :-

a) Ease of use :- Healthcare professionals can easily access patient information without needing technical knowledge.

b) Improved security :- Only authorized users can access the sensitive medical records.

c) Better Data Management :- Large amounts of medical data can be handled efficiently.

d) Reduced Errors :- Standarized views reduce the chances of incorrect data interpretation.

Challenges :-

a) System complexity :- Designing abstraction layers requires careful planning and advanced system design.

b) Implementation cost :- Developing and maintaining such systems can be expensive.

c) Performance Issues :- Multiple abstraction layers may sometimes slow down system performance.

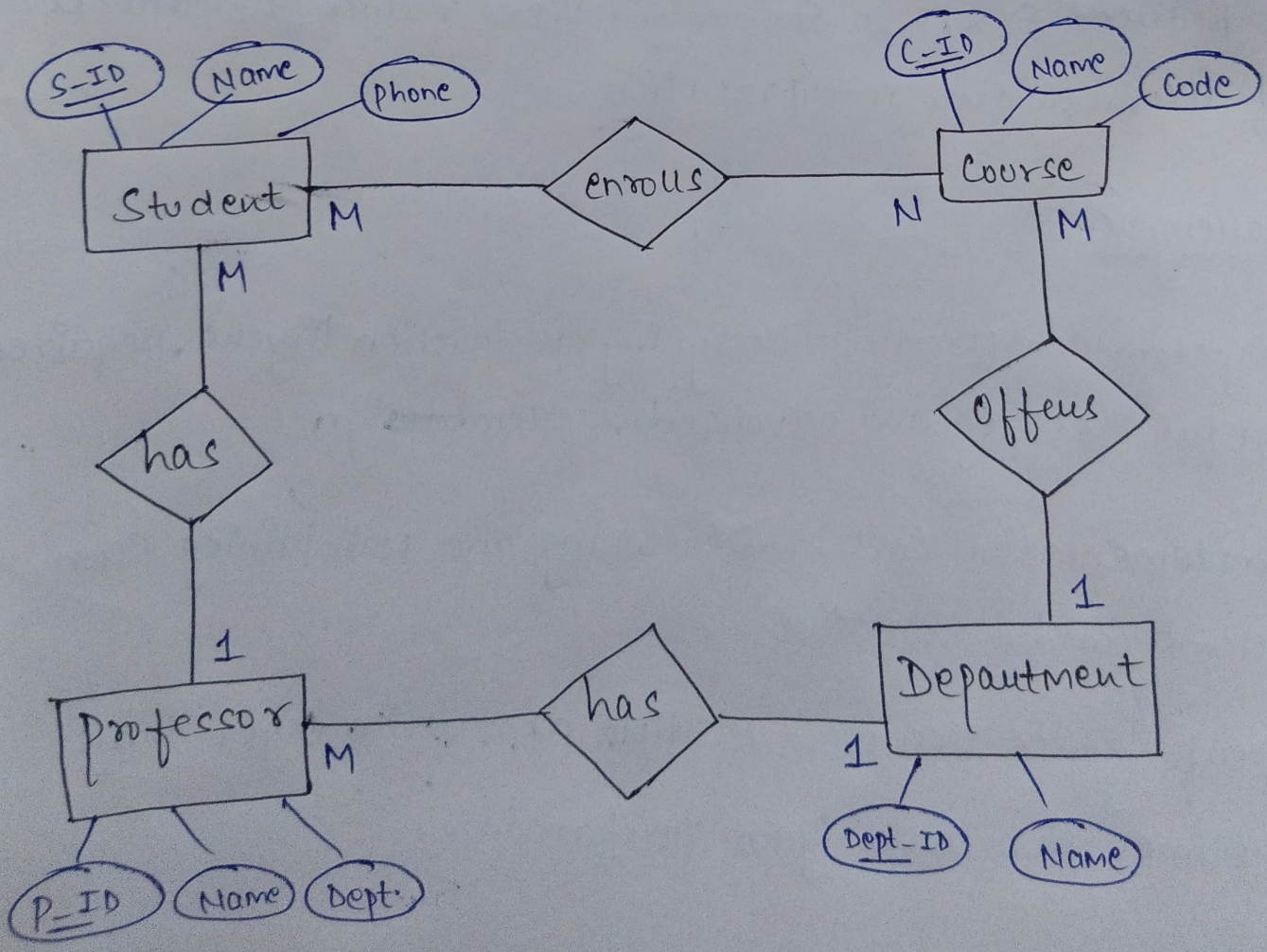
Ques 4 → Design a database for a university using ER model.

Ans → Student (Entity) (S-ID, Name, Phone)
Course (Entity) (C-ID, C-Name, Code)
professor (Entity) (P-ID, Name, Dept)
Department (Entity) (D-ID, Name)

Relationships

Student → enrolls → Course (M:M)

Student → has → professor (M:1)

professor → has → Department (M:1)

Department → offers → Course (1:M)